Chinese Journal of Catalysis 84 (2026) 324–336

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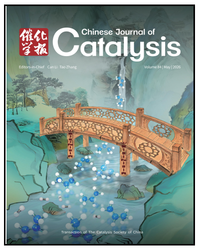

Article

Reactive gas modulation alters metal nanostructures nuclearity and boosts catalytic activity

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A R T I C L E I N F O

Article history:
Received 9 September 2025
Accepted 12 January 2026
Available online 5 May 2026

Keywords:
Single-atomic centers
Metal nanoparticles
Dynamic surface phenomena
Metal nuclearity control
Catalyst activation
Heterogeneous catalysis

A B S T R A C T

This study describes the dynamic behavior of metal nanoparticles on surfaces modulated by reactive gases (CO, NO, H_2, H_2O, and O_2) under soft conditions at low pressure and temperature. Quantum chemical simulations, experimental methods, and machine learning revealed distinct effects: NO promoted nanoparticle fragmentation into highly active single-atom species; H_2, H_2O, and O_2 induced nanoparticle growth; and CO stabilized their structure. This reactive gas modulation (RGM) effect enables flexible control over nanoparticle size and distribution, advancing nanoscale metal tuning. In practical applications, NO gas enhanced the performance of the Pd/C catalyst, facilitating Suzuki-Miyaura cross-coupling under mild conditions (35 °C) with superior efficiency. The developed approach was evaluated for other metals and corresponding effects were studied (Ni, Fe, Co, Cu, Au, Pt, Ru, Ir, Rh), demonstrating versatile possibilities to control nanoscale morphology. The results highlight a flexible metal nuclearity control tool based on the RGM effect in the optimization of catalytic systems for fine organic synthesis, opening the way for advances in catalysis and materials science through nanoscale precision. Through a multilevel study using theoretical and experimental approaches, a methodology for a rapid, energy-efficient and easily scalable approach to synthesize single-atom catalyst at the gram-scale was developed.

1. Introduction

Catalysis is the core basis of modern industry, with up to 80% of all industrially important catalytic processes being carried out via heterogeneous supported catalysts [1]. High durability, low contamination factor of the target product, and ease of separation are among the key advantages of heterogeneous catalysts over their counterpart homogeneous systems [2–6]. Despite these practical advantages, reduced activity is a drawback of many supported systems, especially for fine synthetic transformations in the solution phase. Remarkable C-C cross-coupling reactions have been developed solely on the basis of metal-dissolved homogeneous catalysts and have achieved high activity and selectivity in applications in the fine chemical industry [7,8], pharmaceuticals [9,10], biochemistry [11,12], and agrochemicals [13], among many others. Finding the possibility to improve the activity and match the efficiency of homogeneous catalysts, upon combining the advantages of supported systems, is a key goal of heterogeneous catalysis in solution-phase fine synthetic applications.

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https://doi.org/10.1016/S1872-2067(26)65004-8

In supported systems, the catalytic activity is governed by nano- or sub-nanosized particles of a metal, and the support acts as a stabilizer of these particles [14,15]. In synthetic transformations in solution, catalysis involving metal nanoparticles on the surface of the support is less efficient than in homogeneous systems with similar metal contents [16]. The efficiency of heterogeneous single-atom catalysts (SACs) is considered the closest to that of molecular catalysis and may even surpass that of other methods [17–22]. The key role of palladium SAC centers in cross-coupling reactions has recently been shown [23], and the wide applicability of these catalysts in such demanding areas as fine organic synthesis has been discussed [18,24,25]. Additionally, controlling the state and size of active metal centers on the support is very important for the modern development of new highly efficient electrochemical systems for sustainable development and energy production [26–28].

However, several challenging factors limit practical application of SACs [29]. The synthesis of SACs may require special techniques, and obtaining a pure SAC-only surface organization is difficult (if possible) since SACs/NPs typically coexist in varying ratios [23,30,31].

The second limiting factor concerns the quick degradation of SACs, which is often observed due to catalyst dynamics (Figs. 1(a), (b)). Movement over the support surface leads to SAC capture by metal NPs [23]. In addition, in the catalytic reactions in solution, leaching/redeposition and inevitable agglomeration lead to irreversible conversion of SACs into less active NPs with deterioration of catalyst properties (Fig. 1(b)) [23,32]. Thus, both catalyst changes-surface transfer and coalescence, as well as leaching-induced aggregation-lead to the conversion of SACs to NPs at nearly unfavorable SAC/NP ratios. For Pd systems, upon aggregation into a single NP with a size ranging from 2–3 nm, as many as ~1000 SAC centers may be lost [33–35]. Indeed, the thermodynamic reasons for the formation of NPs from SAC centers and the resistance of NPs to spontaneous spreading back into clusters and atoms include the high strength of the metal-metal bond (90–130 kcal/mol depending on the metal) [36–39].

Since the introduction of the term "single-atom catalysts" in scientific area by Zhang *et al.* [40] in 2011, there has been significant development of this topic. A series of recent studies have demonstrated the importance of maintaining the SAC catalytic framework or restoring SAC centers for efficient processes. Zhang and coworkers [41] demonstrated that in the water-gas shift reaction, the activity of Ir/FeO_x catalysts is dominated by Ir single atoms, which contribute ~70% of the total activity; when Ir atoms agglomerate into clusters or nanoparticles, the specific activity drops by an order of magnitude. Hutchings and coworkers showed that in acetylene hydrochlorination, active particles involve Au_1; when metal atoms combine with nanoparticles, the activity of the system decreases [42]. Treatment with hydrochloric acid leads to the destruction of gold nanoparticles with the release of single-atom centers and revitalization of the activity of the catalytic system [43]. Lu and coworkers [15] conducted a large-scale study of the activity of copper heterogeneous geminal-atom catalysts in C–*X* (*X* = C, N, O, S) bond formation reactions. Cao and coworkers [44,45] demonstrated that the size reduction of copper nanoparticles on dealuminated beta zeolite in methanol vapor was more than twofold, and the advantages of monoatomic copper centers in the methane oxidation reaction. Li and coworkers [46] demonstrated the importance of the nature of the support for a metal catalyst, as it significantly affects the stabilization and dynamic behavior of particles on the surface, using gold and silicon and zirconium oxides as examples. Wang and coworkers [47,48] showed the significance of the processes of metal catalyst dissociation into monometallic particles as well as their agglomeration in multicomponent catalytic reactions for organic synthesis in solution. Leyva-Pérez and coworkers [49,50] revealed the unique properties of SACs based on widely available metals in reactions where precious metals were previously needed. Pagliaro and coworkers [51,52] convincingly demonstrated the trend toward the introduction of single-atom catalysts into industry. Dynamic rearrangement upon interaction with reaction components has been shown [53–55], with examples of reversible destruction of metallic nanoparticles or their spontaneous assembly from surface atoms and clusters [56,57]. The dynamic behavior of metallic nanoparticles in the atmosphere of various gases has

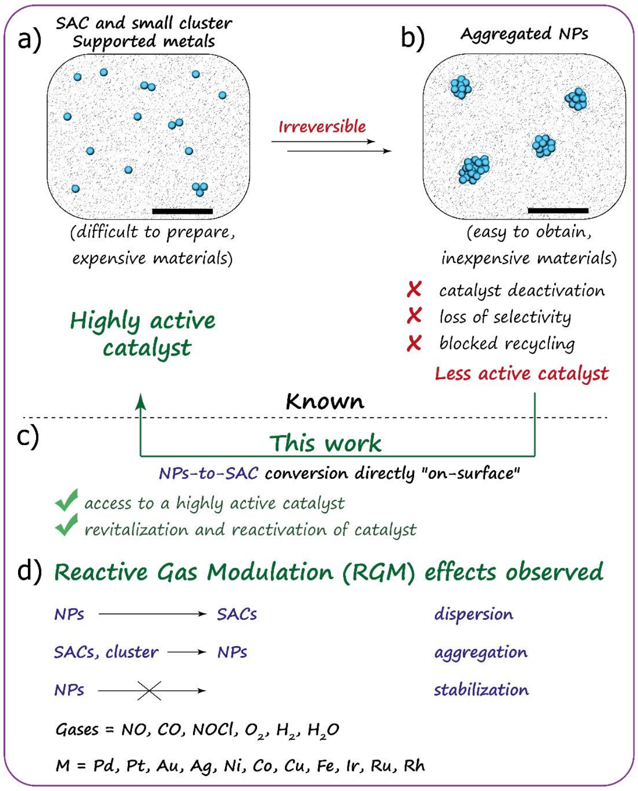

Fig. 1. Known limitations in supported metal particles and reactive gas modulation possibilities described in this work. (a) Subnanometer-sized metal clusters and surface single metal atoms are highly reactive, but the materials are unstable, expensive and difficult to obtain. (b) Under ambient/reaction conditions, small-sized particles can irreversibly aggregate into large crystallites, which are more stable but no longer retain the unique catalytic properties. (c) In this work, we have shown that simple and inexpensive treatment with gas transforms low-activity large metal particles into highly active clusters and atoms. (d) Gas molecules were classified on the basis of their ability to increase/decrease/stabilize nanoparticles of different metals.

been studied, for example, spatiotemporal phase transitions in Pt and Pd nanoparticles [58–60]. Recent studies have shown that the reduction in the size of metal particles require the use of high temperatures, long periods of time, or specialized strongly binding supports to compensate for the energy costs of detaching an atom from a nanoparticle by stabilizing it with a support [61–65]. Modulation of the SAC/NP properties under mild conditions, applicable for typically used metal catalysts for fine organic synthesis in solution, was not yet reported.

Recently, the concept of cocktail-type catalysis was formulated and used to describe different types of metal species (SACs, clusters and NPs) involved in one catalytic system and map their mutual transformations during the reaction [66–68]. For synthetic reactions in solution, bottom-up (SAC-to-NP) and splitting (leaching) transitions may occur and cause major reactivity switches depending on the reaction involved. Such dynamic transitions, rapidly converging into thermodynamically stable but less reactive metal forms, constitute a stumbling block for the application of supported catalysts in solution-state fine organic synthesis. During the precipitation-based synthesis of catalysts, various metal species (*e.g.*, complexes, clusters, and nanoparticles) are formed due to the complex dynamic equilibrium in solvent media. After deposition on a support, a mixture of support-captured metal centers of various types is created – a "cocktail"-type heterogeneous catalyst [69,70]. However, this random composition often lacks the desired catalytic activity since most of the metal may be present in non-optimal forms for the reaction. One promising approach is to develop a method to re-shape this "cocktail" on a surface, which allows for the redistribution of metal between particles of different natures. Convenient and flexible control over the nature of the supported species can be achieved by directing the transformation of supported particles into SAC centers and unifying metal centers on the surface.

In the present study, we describe a simple and practical approach to control interconversions between SACs and NPs (Fig. 1(c)). The coordination of NO, CO, H_2, H_2O and O_2 gases resulted in different reactivity patterns of Pd particles supported on a carbon surface. *Ab initio* simulations validated by electron microscopy and machine learning (ML) revealed the dynamic behavior of palladium nanoparticles on graphite. The influence of gases can increase the size of nanoparticles (H_2O, H_2, O_2), stabilize their size (CO) or lead to their redistribution to smaller palladium particles (NO). The application of NO in catalysis demonstrated the unique ability of NO on Pd/C to increase the catalytic activity in C-C cross-coupling reactions under very mild conditions and with low palladium loading. Depending on the gas treatment applied, the reactive gas modulation effect can be used to initiate dispersion, aggregation or stabilization of particles (Fig. 1(d)), as illustrated for a variety of metals (Ni, Fe, Co, Cu, Ag, Ru, Ir, Pt, Rh and Au). This approach allowed us not only to detect and explain the movement of nanoparticles and metal atoms on the surface but also to develop a simple and fast method for the preparation of catalysts based on single metal atoms in gram quantities in a short time and with lower energy consumption than previously developed methods.

2. Experimental

2.1. Pd/C sample preparation

4 mL of chloroform was poured into the vial, and 0.4 mg of $Pd_2dba_3{\cdot}CHCl_3$ was added. A graphite bar (25×5×2 mm) was placed in the resulting solution, and the mixture was heated at 75 °C until bleaching. After the precipitation of palladium, the resulting catalyst was washed with acetone and dried under vacuum. The complex ($Pd_2dba_3{\cdot}CHCl_3$) was obtained according to a published methodology [71].

2.2. Characterization

The sample morphology was studied via a Regulus8230 (Hitachi) scanning electron microscope (SEM) at an accelerating voltage of 20 kV. Electron microscopy at identical locations was performed according to published guidelines [23]. The sample for transmission electron microscopy (TEM) was prepared in air by grinding the powder in an agate mortar in dry ethanol and depositing drops of suspension onto a copper TEM grid with a holey carbon layer. High-angle annular dark field scanning transmission electron microscopy (HAADF-STEM) images and energy-dispersive X-ray (EDX) spectra were acquired with a probe-corrected Titan Themis Z transmission electron microscope operated at 200 kV and equipped with a Super-X EDX detector.

The X-ray photoelectron spectroscopy (XPS) measurements of Pd/C were carried out by using an OMICRON ESCA+ system at an 8×10^{-10} mbar pressure and 20 eV transmission energy, with the Mg anode as the source of radiation at a 1253.6 eV radiation energy and a power of 252 W. The positions of the Pd 3*d* XPS peaks were standardized to the C 1*s* peak with a binding energy of 285.0 eV. The spectra were analyzed after subtraction of the background determined via the Shirley method [72]. The positions of the peaks were determined with an accuracy of 0.1 eV. The element ratios were calculated from the integral intensities under the peaks, taking into account the photoionization cross sections σ of the corresponding electron shells [73].

2.3. General procedure of the gas treatment of Pd/C

Palladium on a graphite bar was placed in a glass test tube heated to 100 °C. Then, the following gases – NO, H_2, and O_2 in 2 liters of each gas were passed through the test tube for 3 h, in the case of H_2O, 100 mL of liquid water evaporated. To ensure the purity of the experiment and confirm the absence of dynamic behavior of the nanoparticles on the surface, 4 liters of CO gas were passed through the test tube for 6 h. CO was obtained by reacting concentrated sulfuric acid with formic acid under heating. The gas was then passed through a layer of $CaCl_2$ for drying. H_2 was obtained by the reaction of zinc with dilute sulfuric acid. The gas was passed through a layer of $CaCl_2$ for drying. NO was obtained by the reaction of copper with dilute nitric acid (~30%). Before starting the experiment, the entire setup was purged with argon to displace the oxygen. For dehydration, the gas was passed through a layer of $CaCl_2$. In addition, a comparison was made with commercial NO of 99.5% purity at room temperature and 50 atm. O_2 high-purity oxygen (99.9999%) was purchased and supplied without prior dehydration. H_2O vapor obtained by boiling distilled water was passed through the system without pretreatment.

2.4. Pd/C sample preparation for the catalytic experiments

15 mL of chloroform was poured into the vial, and 10.0 mg of $Pd_2dba_3{\cdot}CHCl_3$ was added. Then, 2.00 g of graphite powder (Sigma Aldrich, < 20 μm, synthetic) was added to the resulting solution, and the mixture was heated at 75 °C until bleaching (approximately 5 min) [74]. After the precipitation of palladium, the resulting catalyst was washed with acetone and dried under vacuum. The mass fraction of palladium in the obtained catalyst was 0.1 wt%. Half of the obtained catalyst (1.00 g) was treated with NO gas according to the general procedure. A special comparison of the processing time and processing temperature by gaseous NO showed that 30 min at room temperature was sufficient for effective activation of the heterogeneous Pd/C catalyst.

2.5. General methodology for catalytic reactions

Aryl bromide (0.5 mmol) and aryl boronic acid (0.5 mmol) were dissolved in 1.6 mL of N-methyl-2-pyrrolidone (NMP). K_2CO_3 (82.8 mg, 0.6 mmol) was dissolved in 0.4 mL of water (see Table S1 in Supporting Information for more details). The resulting solutions in NMP and water were mixed in a previously unused test tube with a new magnetic stirring anchor. A catalyst (0.1 wt% Pd/C) of 0.2 mol% palladium relative to aryl bromide was then added. The reaction mixture was left under stirring for 12 h at 35 °C. ^{1}H nuclear magnetic resonance (NMR) spectroscopy was used to analyze the reaction mixture (Bruker Fourier 300 HD spectrometer, frequency of 300.1 MHz (^{1}H)). The residual solvent peak (DMSO-d_6 2.50 ppm) was used as an internal standard for ^{1}H. For this purpose, a 100 μL sample of the reaction mixture was placed in an NMR tube, and 500 μL of deuterated solvent (DMSO-d_6) was added. The presence of the Suzuki-Miyaura coupling reaction product was also confirmed by GC-MS.

2.6. Computational details

Simulations of the palladium atoms and nanoparticles on the graphene surface were carried out within the framework of spin-polarized density functional theory (DFT) as implemented in VASP [75–77]. The revised Perdew-Burke-Ernzerhof parametrization (rPBE) [78] of the exchange-correlation functional was used. The plane-wave energy cutoff was set to 600 eV. To treat the weak van der Waals interaction, we used the DFT-D3 method of Grimme *et al*. [79] for the dispersion energy-correction term.

The palladium atom placed on the graphene sheet with 50 atoms in the unit cell with parameters of $a = b = 12.4$ Å was simulated to analyze the electron density redistributions, chemical bonding, and energy barriers. The Pd nanoparticle contains 38 atoms and was located on the graphene with a unit cell of 240 atoms and cell parameters of $a = b = 24.81$ Å to vanish the interaction between neighboring nanoparticles. For each Pd nanoparticle, the maximum number of adsorbed molecules was 25, representing full coverage of the nanoparticle.

The Monkhorst-Pack [80], k-point grid of 2×2×1 was used to sample the first Brillouin zone of the graphene with isolated Pd atoms and Pd atoms with adsorbed molecules. For the relaxation and investigation of the system containing the graphene sheet and Pd_{38} cluster, only Γ-point calculations were used. Relaxation of the simulated structures was carried out until the total energy difference and the atomic net forces became less than 10^{-5} eV and 10^{-4} eV/Å, respectively, to ensure high accuracy and convergence of the calculations. The atomic structures and charge densities were illustrated using the VESTA package [81]. The diffusion barriers for Pd atoms and Pd nanoparticle on graphene were examined using the climbing image nudged elastic band (CI-NEB) method [82], with 28 images for Pd atom and 14 images for Pd nanoparticles constructed for each diffusion pathway.

The crystal orbital Hamilton population (COHP) method [83] was used to study the chemical bonding between Pd atom and graphene. To extract information about chemical bonding from the electronic density of states (DOS) of a system, the crystal orbital overlap population (COOP, population of mutually overlapping crystal orbitals) was used. COOPs are obtained by multiplying the DOS by the number (population) of overlaps. This makes it possible to separate the total electron density into bonding and anti-bonding states. Therefore, the theory of the crystal orbital Hamilton population (COHP) was proposed that created populations of overlapping orbitals. The sign of the COHP is opposite to that of the COOP in the case of the bonding and anti-bonding states. Therefore, -COHP diagrams are always built for clarity. COHP allows one to obtain information about bonding and anti-bonding states in a simple manner. The mathematical form of the COHP is very similar to that of a projected DOS equation, but the density-of-state matrix is weighted by the Hamilton matrix elements. A periodic crystal can be defined as follows [84]:

$$COHP_{AB}(\epsilon) = \sum_{\mu\in A}\sum_{\nu\in B} COHP_{\mu\nu}(\epsilon) \tag{1}$$

$$COHP_{AB}(\epsilon) = \frac{2}{V_{BZ}}\int_{BZ} H_{\mu\nu}(k)\left\{\sum_j c^*_{\mu j}(k)c_{\nu j}(k)\delta[\epsilon-\epsilon_j(k)]\right\}dk \tag{2}$$

where $H_{\mu\nu}(k)$ is the Hamiltonian matrix in reciprocal space and where $c_{\nu j}(k)$ are the expansion coefficients of the *j*-th occupied crystal orbital with energy $\epsilon_j(k)$ in terms of Bloch functions numbered in the same way, such as atomic orbitals, by the subscript μ or ν. Integration over k in Eq. (2) is performed in the Brillouin zone (BZ) with a volume of V_{BZ}.

Owing to weighting with the elements of the Hamiltonian matrix, the bonding and anti-bonding interactions are indicated by negative and positive COHP values, respectively. Integration of the COHP up to the Fermi level (E_F) yields integrated COHP (ICOHP) values:

$$ICOHP_{AB} = \int_{E_F} COHP_{AB}(\epsilon)d\epsilon \tag{3}$$

COHP calculations were carried out using the LOBSTER algorithm [85], and further consideration of the contribution of each orbital to the total binding energy was carried out using the Dragon software package.

3. Results and discussion

To study the phenomenon of the migration of Pd species on the surface caused by interactions with gas molecules, we performed simulations of the energy barriers of the migration of isolated Pd atom on the graphene surface. Climbing image nudged elastic band (CI-NEB) calculations were performed to study the migration of an isolated Pd atom and a Pd atom with adsorbed CO, NO, H_2, H_2O, and O_2 molecules on the graphene surface, as shown in Figs. 2(a) and (b). For the Pd atom with adsorbed CO, NO and H_2, the most favorable position of the molecules on the graphene is located on top of the C atom (Fig. 2(c)). For Pd atom with adsorbed H_2O and O_2 molecules and for

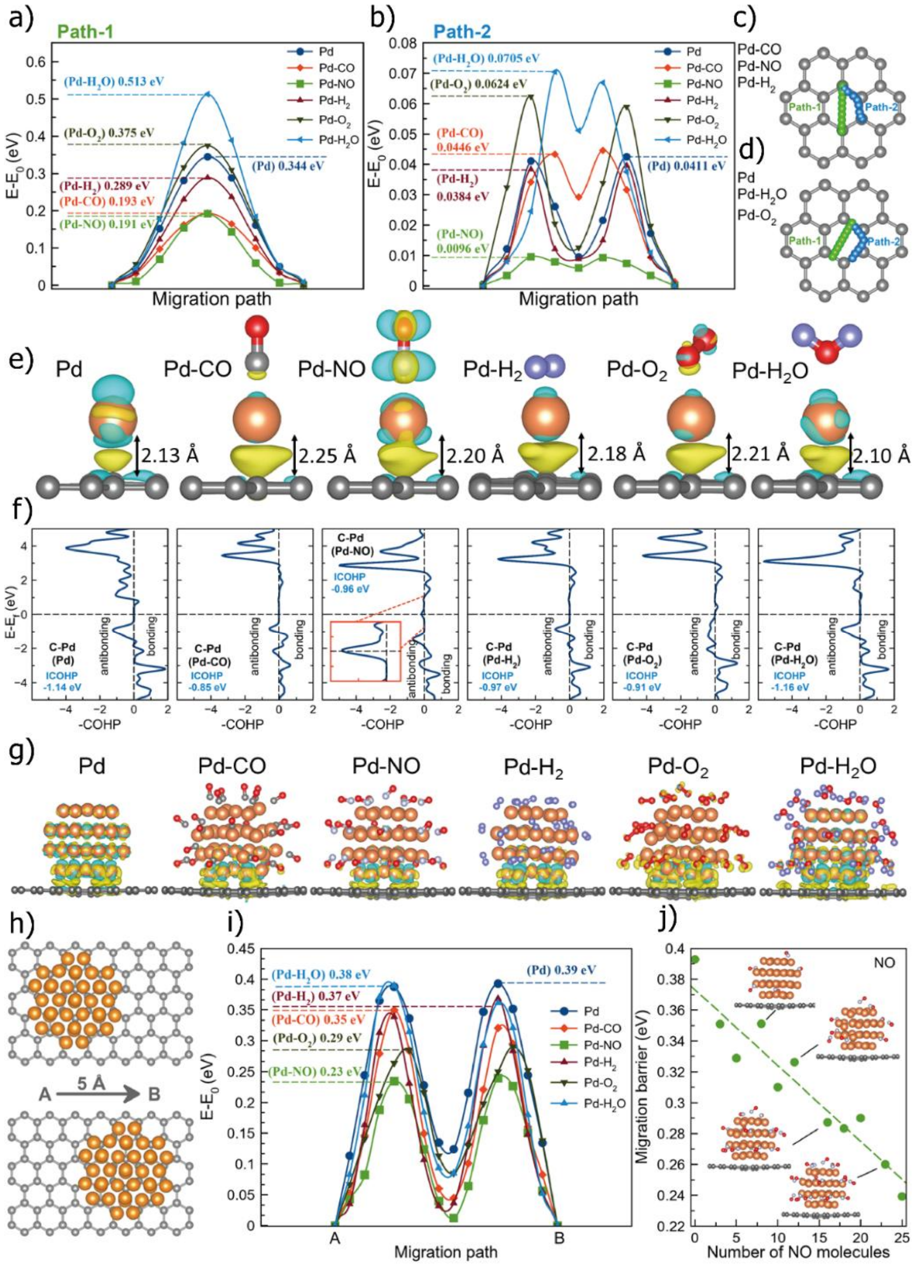

Fig. 2. Computational modeling at the atomic level. Calculated migration barriers for Pd atoms with adsorbed molecules for Path-1 (a) and Path-2 (b). Schematic illustration of Path-1 and Path-2 for adsorbed NO, CO, H_2 (c), and O_2, H_2O and pure Pd (d). Path-1 is shown in green, and Path-2 is shown in blue. (e) Visualized charge redistribution for Pd atoms with adsorbed molecules on graphene. The yellow color of the density corresponds to an excess of electrons, whereas the blue color corresponds to electron deficiency. The iso-value is 0.025 e/Å^3. (f) Calculated COHP diagrams for the studied systems. (g) Atomic structures of Pd_{38} nanoparticles with adsorbed CO, NO, H_2, O_2, and H_2O molecules on the graphene surface together with charge redistribution between graphene and Pd_{38}. The yellow color of the density corresponds to an excess of electrons, whereas the blue color corresponds to electron deficiency. The iso-value is 0.01 e/Å^3. (h) Illustration of the migration pathway of Pd_{38} nanoparticles on graphene. (i) Calculated migration barriers for Pd_{38} nanoparticles over graphene. (j) Dependence of the migration barrier on the NO concentration adsorbed on the Pd_{38} nanoparticle.

pure Pd atom the most favorable position on graphene is on top of the C–C bond (Fig. 2(d)). Therefore, the initial positions for calculating the migration barrier of Pd atoms with different adsorbed molecules are chosen according to their energetic favorability.

Two migration pathways were considered along which the Pd atom can migrate over the graphene surface, called as Path-1 and Path-2 (Figs. 2(c) and (d)). Path-1 involves the migration of Pd atom through the graphene hexagon (green color in Figs. 2(c) and (d)), whereas Path-2 involves the migration around the hexagon (C-C-C fragment of the hexagon, blue color in Figs. 2(c) and (d)). Interestingly, the calculated migration barrier for Path-1 for all the molecules is approximately 10 times greater than that for Path-2 (Figs. 2(a) and (b)). The lowest value is for the Pd atom with the adsorbed NO molecule (0.191 eV). The migration barrier for Path-2 is much smaller than that for Path-1, namely, the Pd atom with adsorbed NO molecules has the lowest barrier of 0.0096 eV (Fig. 2(b)). The reduction of the migration barrier of the palladium atom on graphene from 0.0411 to 0.0096 eV in the presence of the sorbed NO molecule is equivalent to a 360 K decrease in the displacement onset temperature from +200 to −160 °C. On the other hand, the effect of H_2O raises the migration temperature to +550 °C.

To understand this difference in the migration barriers of the Pd atom with different molecules, we calculated the electron density redistribution between graphene and the Pd atom with different adsorbed molecules, as shown in Fig. 2(e). We calculated the charge redistribution for different multiplicities of the considered molecules (see Supporting Information for more details). In all the cases, the greatest charge density redistribution is observed in the C–Pd bond, but it is not clear how it affects the strength of the C–Pd bond. Visually, a greater redistribution was observed for the Pd-NO case. To study the C–Pd bond in more detail, we calculated the Crystal Orbital Hamilton Populations (COHPs). The calculated −COHP diagrams are shown in Fig. 2(f). For all the considered cases, there are antibonding states at the Fermi level only in the case of Pd-NO, which reduces the bonding of graphene with the Pd-NO complex, decreasing the migration barrier compared with that of other molecules (Fig. 2(f)).

Next, we studied the migration of Pd_{38} nanoparticle with different adsorbed molecules on the graphene surface. Actual particle motion can be very complex, combining elements such as sliding, rolling, and rotation. In our study, we focused on the sliding (translational motion) of nanoparticles on the support surface, as previously described by Wang and coworkers [86]. Here, we consider a larger graphene cell (25×25 Å) to exclude the interaction between periodically arranged nanoparticles. For each type of gas molecule, the number of adsorbed molecules on the Pd_{38} surface is equal to 25. The atomic structure of Pd_{38} with adsorbed molecules with visualized charge redistribution between graphene and the nanoparticles is shown in Fig. 2(g). This allows us to simulate the situation in which the Pd_{38} particle is in the saturated atmosphere of the selected gas.

We have considered the migration pathway along the zigzag direction of graphene, as shown in Fig. 2(h). The migration distance is 5 Å. The results of the calculation of the migration of Pd_{38} over the graphene surface showed that, as in the case of Pd atoms with adsorbed NO, Pd_{38} with adsorbed NO molecules also has the lowest migration barrier (see the green color in Fig. 2(i)). The migration barrier for pure Pd_{38} particle is 0.39 eV, but in the presence of defects on the carbon surface it can increase to 3 eV (Figs. S28–S30), whereas for Pd_{38}, the barrier for adsorbed NO molecules is reduced to 0.23 eV. Considering that the case of 25 adsorbed molecules is a limited case, we have studied how the migration barrier changes with the number of NO molecules adsorbed on the surface of Pd_{38}. It is observed that migration barrier increases almost linearly as the number of NO molecules adsorbed on the Pd_{38} nanoparticle decreases, as shown in Fig. 2(j). The presence of NO not only lowers the barriers to movement across the surface of defect-free graphene but also significantly reduces the stabilization force of such a strongly binding defect as a monovacancy (Figs. S34, S35). Doubly coordinated metal atoms, which can form with a large excess of NO, are almost completely unstabilized by the surface defect (Fig. S36), explaining the observed movement of atoms over large distances from the initial nanoparticle.

To determine the influence of gases on Pd catalysts, we studied how different molecules adsorbed onto Pd surfaces (111) and (100) change the bonding between adsorbed molecules and neighboring Pd atoms (see highlighted atoms in Figs. 3(a) and (b)). For this purpose, we calculated the integrated COHP (ICOHP), which reflects the bond strength (the more negative the value is, the stronger the bond strength) (Fig.

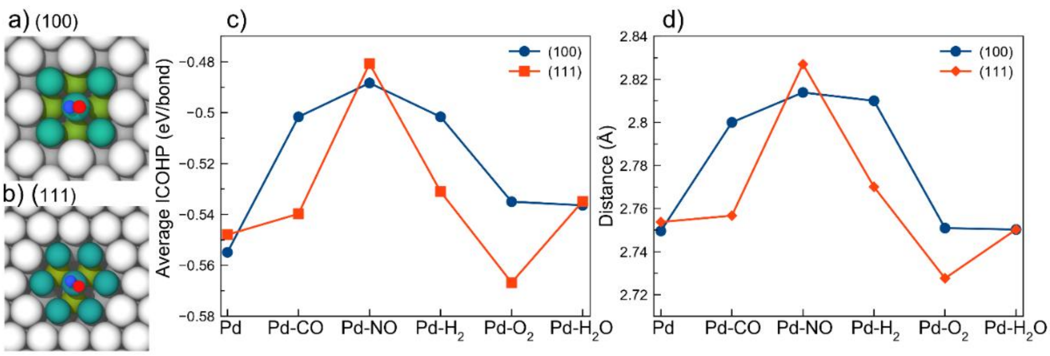

Fig. 3. Modeling the local environment effect. (a,b) Illustration of the local environment of the (100) (c) and (111) (d) Pd surfaces for which the ICOHP was calculated. Dependence of integrated COHP (a) and bond distance (b) on the type of adsorbed molecule.

3(c)). In the case of NO, the value of ICOHP is the largest, which indicates the weakening of the bonding between the Pd atom with adsorbed NO and the neighboring Pd atoms (Fig. 3(c)). The same tendency is observed for both the (100) and (111) surfaces of Pd. Weakening of the chemical bonding increases the bond length (average value of the bond length with neighboring Pd atoms) (Fig. 3(d)).

Theoretical analysis of Pd/C systems in various gas atmospheres has shown prerequisites for the dynamic behavior of surface nanoparticles. Of particular interest is the discovery of the disintegrating (dispersion) effect of NO on nanoparticles, which could open an easy pathway for the conversion of NPs-to-SAC. Therefore, the next step was to experimentally investigate the effect of gases on Pd/C.

To validate the theoretical findings, an experimental study of Pd/C systems in different atmospheres was performed, and the results were analyzed with ML algorithms. The detailed experimental conditions are given in the Experimental part. The behavior and dynamics of specific nanoparticles were investigated via the 4D-catalysis approach [67], which allows the tracking of specific surface areas of the catalyst before and after gas treatment with nanometer precision. Changes in the size and mutual arrangement of the nanoparticles before and after gas treatment were carried out via neural networks (Fig. 4).

To analyze the nanoparticle shifts before and after the reactions, we used the following approach (Fig. 4(a)). Initially, we identify keypoints in each image to be matched; then, we match the corresponding features; and for the top 50 matches, we align the images via a homography transform. The same transformation is also applied to the neural network output segmentation masks. These aligned masks are then processed (using a combination of the watershed algorithm and additional postprocessing), and absolute within-frame nanoparticle positions are compared. This is done by calculating distances to the nearest neighbors in the frame from the other image. An example of a matched nanoparticle is shown in Fig. 4(b), and the results of the analysis are shown in Fig. 4(f), where after the reaction sizes of the nanoparticles increase, no significant correlation with the magnitude of the shift is observed. Furthermore, the images can be analyzed directly, using embeddings from a pretrained ResNet model, and every process happening to the samples can be analyzed as a shift in the vector space

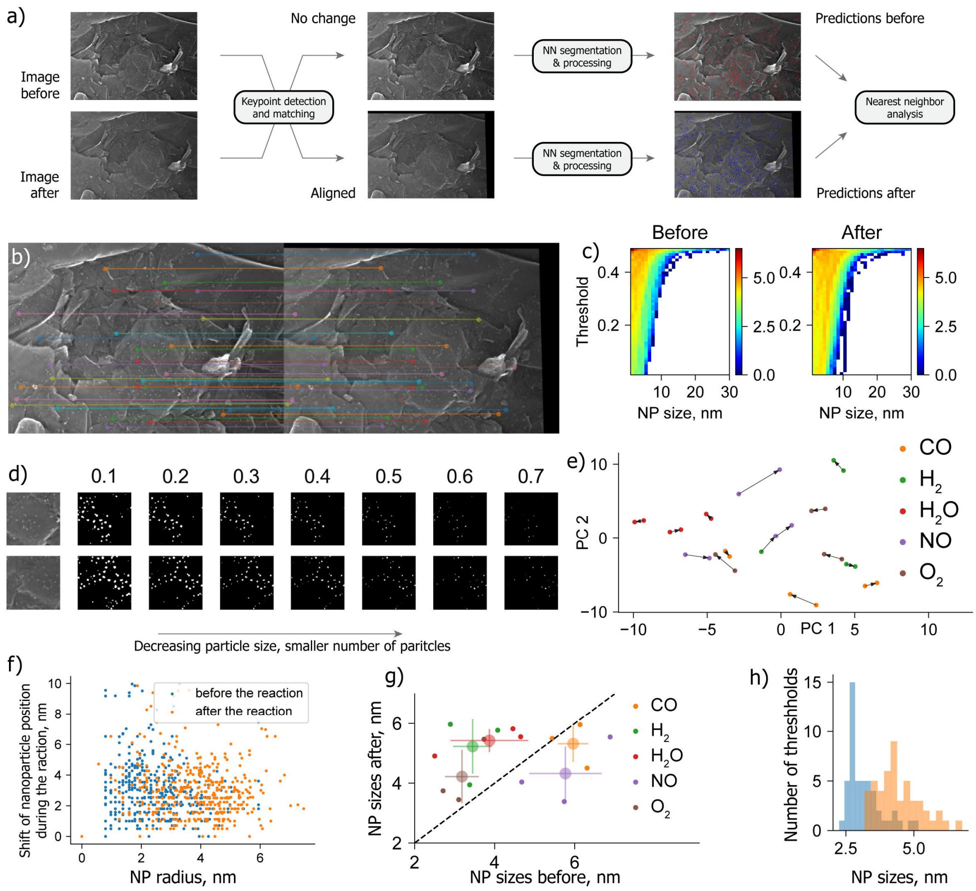

Fig. 4. Computational analysis pipeline for palladium nanoparticles on graphite. An automated algorithm for analyzing images obtained with an electron microscope is exemplified with an argon atmosphere. A side-by-side comparison of all images for CO, H_2, H_2O, NO and O_2 is presented in the supplementary information. (a) Overview of the pipeline, including image matching and neural network analysis steps. (b) Examples of two aligned images and matched nanoparticles. (c) Distribution of nanoparticle sizes at different thresholds, showing an average increase in nanoparticle size across the entire threshold range. (d) Relationship between the binarized neural network output and threshold, highlighting common problems in nanoparticle size determination. (e) Dimensionality reduction plot showing how images move within the image vector space during the reaction. (f) Comparison of nanoparticle radii and shifts in two directions. (g) Analysis of nanoparticle distribution throughout the dataset; error bars indicate standard deviations along each axis. (h) Distribution of median sizes across all thresholds in the example images from 'c'.

(Fig. 4(e)). For example, it can be seen that, on average, the images move toward 0 in picture 1 (PC1). Moreover, depending on the gas, the magnitude of this shift varies—small for water but large for nitrogen monoxide and oxygen. Additionally, we wanted to obtain a better estimation of the nanoparticle sizes. Unfortunately, this number is highly dependent on the threshold, as nanoparticle sizes are relatively small compared with those of the remaining items in the images, and the collected data exhibit several artifacts (Fig. 4(d)). To solve this problem, we first attempted to analyze the distribution of nanoparticle sizes (obtained through the same procedure as described before) across a wide set of thresholds (Fig. 4(c)) to binarize segmentation masks before applying the watershed algorithm (generally, lower thresholds increase the particle size, whereas the relationship with the number of particles is nonlinear). We then calculated distributions of median sizes across each threshold (Fig. 4(h)). The means of these distributions were then used to characterize the average apparent size of each nanoparticle in the images before and after the reaction, as shown in Fig. 4(g). As one can see, only for NO do we observe a significant apparent reduction in nanoparticle sizes; CO remains mostly unchanged, and other gases show an increase in nanoparticle sizes.

Thus, the experimental study of the influence of a number of reactive gases on the behavior of heterogeneous catalysts of Pd/C species revealed a dynamic character in systems with H_2, H_2O, O_2 and NO, as predicted above by *Ab initio* calculations. A change in the size of the nanoparticles dramatically altered the catalytic activity of the system. This fact is important because the processes occur directly with active centers. We then investigated in detail the catalytic potential of nanoparticle reshaping under the action of gaseous NO.

The discovered effect of NO-induced destruction of palladium nanoparticles has shown powerful practical application potential. Indeed, in reactions involving C–C bond formation, such as Mizoroki-Heck and Suzuki-Miyaura cross-coupling, single palladium atoms show the highest activity [23,87]. To investigate the synthetic potential of the observed effect on Pd/C catalysts, we performed the following experiments. The catalytic activity of a Pd/C (0.1 wt% Pd) sample divided into 2 parts, one treated with NO (Pd/C^{NO}) and the other left as Pd/C (Fig. 5(a)), was tested in Suzuki-Miyaura reactions. The NO gas used for palladium atomization was passed through a flask containing water at the reactor outlet. This solution was then analyzed for palladium content using inductively coupled plasma atomic emission spectroscopy (ICP-AES). No palladium was detected, indicating no palladium loss during processing. IR analysis of residual NO using the diffuse reflectance infrared Fourier transform (NO-DRIFT) method did not detect sorbed NO on Pd (Fig. S17) [88]. The CO-DRIFT method revealed changes in the IR absorption spectrum of CO on palladium for both untreated and NO-treated catalysts. The absorption pattern shows that metallic palladium is initially present; however, after treatment with NO, the metallic phase virtually ceases to produce a signal, and a linear Pd^{2+}-CO absorption component appears (Fig. S18) [89]. The original catalyst Pd/C contained palladium nanoparticles, and no single atoms were detected on the surface using HAADF-STEM (Fig. 5(b)). A detailed observation of the surface of the Pd/C^{NO} catalyst using aberration-corrected scanning transmission electron microscopy revealed a high degree of atomization of the original nanoparticles. Palladium has been converted to isolated atoms, and some small clusters of the metal stabilized by the support are also present (Fig. 5(c)).

Interestingly, the destruction of palladium nanoparticles by NO was also observed for another type of support. In the initial form of the Pd/SiO_2 catalyst, only metal nanoparticles were observed (Fig. 5(d)). After exposure to NO, the SiO_2 surface contained a large number of single atoms, as well as small clusters and residual nanoparticles (Fig. 5(e)). Of course, degree of atomization should depend on the properties of the support and on the metal-support interaction. As evidenced from the experiment, the atomization effect was clearly observed for carbon and SiO_2 supports. After the treatment, the catalytic properties of the material change significantly.

The NO gas-treated catalyst was found to have high catalytic activity, allowing the Suzuki-Miyaura catalytic reaction to proceed under very mild conditions at 35 °C and showing a definite advantage in terms of reaction completeness over the untreated catalyst even for deactivated aryl bromides, such as 1a and 1b (Fig. 5(f)). This reaction temperature was chosen as the optimum between activity and best solubility of all substrates in the reaction system. Some substrates show a very strong increase in Pd/C^{NO} catalytic activity, such as 1b, 1c and 1d, where few or no reactions occurred on Pd/C. Polyaromatic bromides 1e and 1f are less sensitive to the NO-treated catalyst, but the completeness of the reaction is still greater on Pd/C^{NO}. Activated substrates 1g–k can almost completely form the product on Pd/C^{NO}, whereas on Pd/C, the reaction did not proceed even half or one-fifth (1g) at the same time.

The study of the kinetic dependences on the example of the synthesis of 4-nitro-1,1'-biphenyl 1i showed that the rate constant of the reaction catalyzed by Pd/C^{NO} is 3 times greater than that of the initial catalyst (Fig. 5(g)). For the unmodified catalyst, the induction time was also approximately 30 min. During this time, no products were detected in the reaction medium. However, the NO-treated catalyst shows the immediate appearance of a product signal, and the induction time, if present, is negligible. A separate quantitative analysis of the leached palladium showed that within 30 min of the reaction's onset, 0.84% of the palladium had been released into solution in the Pd/C system, while 1.58% had been released in the Pd/C^{NO} system. After 12 h, these values were 3.35% and 4.75%, respectively. This result demonstrates that it is not the total palladium content in the solution that is important, but rather the form in which it is present. This demonstrates a bright increase in the efficiency of single-atom catalysis, with an increase in leaching of less than 1.5%.

Various types of supports for palladium, including SiO_2 (see the Supporting Information for more details), showed an increase in activity after exposure to NO in all the samples. Thus, NO treatment of the Pd catalyst improves its performance so that catalytic reactions can proceed under very mild conditions. The comparison of Pd/C^{NO} activity with the catalytic activity of

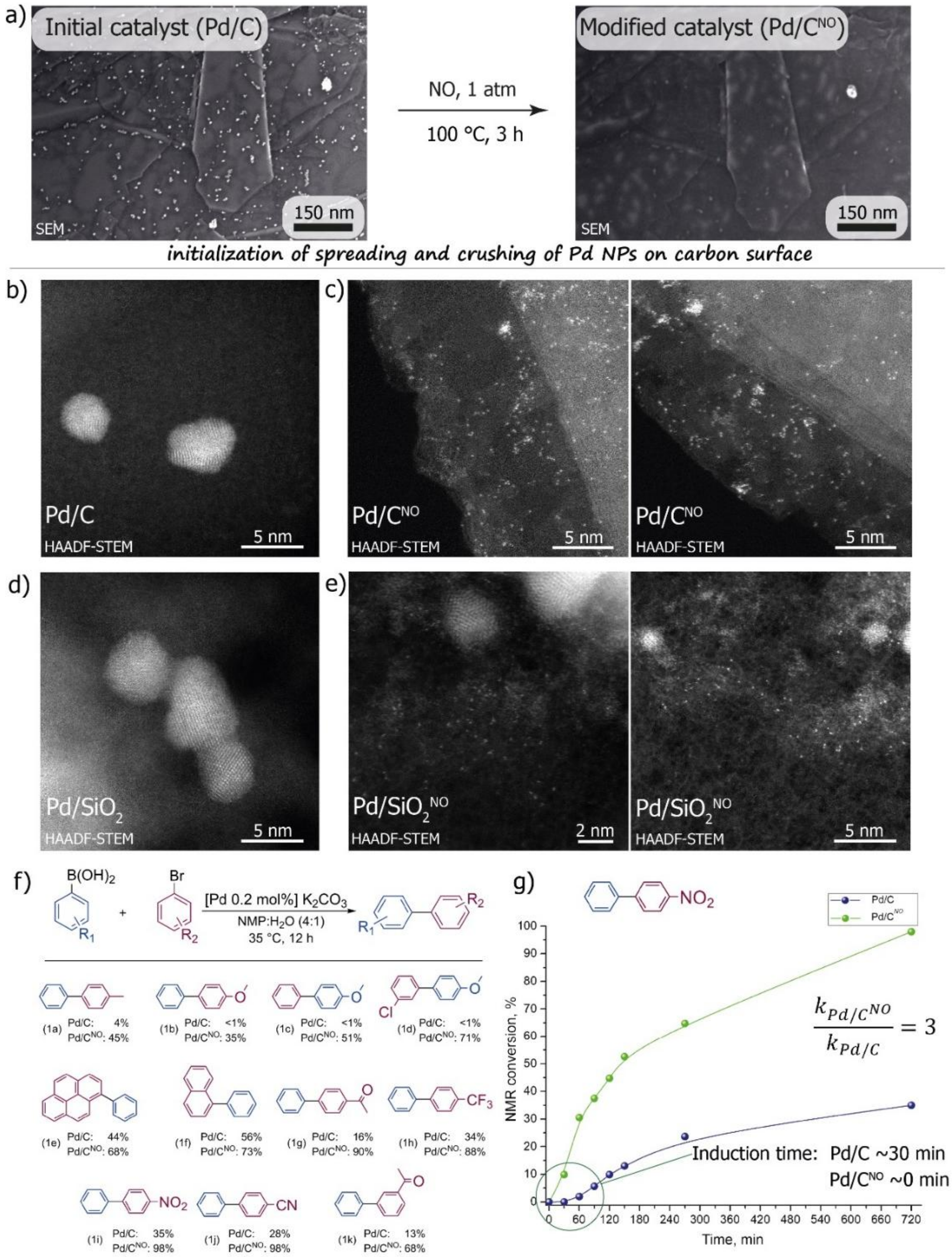

Fig. 5 Catalytic experiments comparing the activity of the initial and gaseous NO-treated Pd/C catalysts. (a) Modification of a nanoscale catalyst by treatment with gaseous NO. (b) HAADF-STEM image of initial Pd/C (C – graphite); Pd single atoms was not found. (c) HAADF-STEM images of Pd/C^{NO} which shows a large number of single palladium atoms as well as the presence of small clusters. (d) HAADF-STEM image of initial Pd/SiO_2; Pd single atoms was not found. (e) HAADF-STEM images of Pd/SiO_2^{NO} which shows a large number of single palladium atoms as well as the presence of small clusters. HAADF-STEM images before and after NO treatment were obtained from different areas. (f) Scope and yield comparison of the products of Suzuki-Miyaura cross-coupling reactions catalyzed by Pd/C systems with/without NO treatment. The catalytic activity results revealed an increase in the activity of the NO-treated catalyst in the C–C bond formation reaction. (g) Study of reaction kinetics with Pd/C and Pd/C^{NO} catalysts on the example of 4-nitro-1,1'-biphenyl synthesis.

homogeneous catalysis on $Pd(OAc)_2$ showed similar activity (Table S7), which indeed reflects the possibility of simple and rapid preparation of the heterogeneous catalyst proposed in this study, whose activity is close to or the same as that of the homogeneous catalyst.

The effect of particle growth is another interesting aspect of the discovered dynamic behavior of gases on the surface of palladium particles. In hydrogenation reactions with molecular hydrogen, the catalysis efficiency depends significantly on the size of the palladium nanoparticles [90,91]. An increase in the size of the metal particles led to an increase in the catalytic activity of the Pd/C system. Therefore, we performed a nanoparticle crushing experiment under NO (Pd/C^{NO}) followed by the initial growth of previously crushed palladium particles under hydrogen gas at high pressure (Pd/C$^{NO\text{-}H2}$). A comparison of the catalytic activities of native Pd/C, Pd/C^{NO} and Pd/C$^{NO\text{-}H2}$

in the hydrogenation of diphenylacetylene with molecular hydrogen at 70 atm revealed that the conversions of diphenylacetylene to cis-stilbene were 62%, 1% and 11%, respectively. To remove possible traces of sorbed NO on the palladium surface before hydrogen treatment, the sample was heated under vacuum, and the conversion of the hydrogenation reaction on such a catalyst was 3% (Table S10). The destructive effect of NO is likely more pronounced than the collecting effect of H_2, so the complete restoration of catalytic efficiency does not occur. These practical effects require special optimization of conditions for a particular hydrogenation reaction and show the fundamental possibility of easily adjusting the size and activity of Pd/C systems. The proposed method for the preparation of Pd SACs is easily scalable by using a 50 atm reactor, which allows the preparation of gram quantities of catalyst in 1 h at room temperature (see the Supporting Information for details and Fig. S49).

We demonstrate the versatility and ease of sputtering many metallic nanoparticles other than palladium. The action of nitrogen monoxide at room temperature for 1 h has been demonstrated on nanoparticles of metals such as Ni, Fe, Co, Cu, Ag, Ru, Ir, Pt, Rh and Au predeposited on graphite (Fig. 6 and Supporting Information). An image comparison of the M/C and M/C^{NO} samples revealed that the Ni, Fe, Cu and Co nanoparticles are sensitive to the action of NO and disperse on the graphite surface (Type I metals). However, interestingly, Au and Ag also have a dynamic nature but are of a different kind (Type II). Unlike Type I metals, here, there is no dispersion of nanoparticles on the surface; instead, some Ag and Au NPs fuse with each other and form from smaller particles on the graphite surface. Fig. 6(a) clearly shows that new gold nanoparticles are formed from clusters of very small particles. Moreover, silver nanoparticles start to actively move across the surface for distances of up to several hundred nanometers, some of the particles grow, and some break up into several smaller particles that merge into small clusters of 2–3 nanoparticles. The metals Pt, Ir, Rh, and Ru do not have any activity toward nitrogen monoxide, so

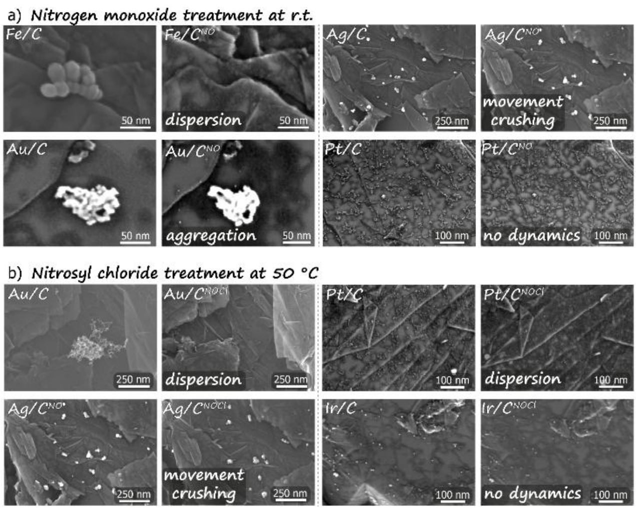

Fig. 6. Surface dynamic behavior. Atomization/dispersion, motion and disintegration of various metallic nanoparticles in an atmosphere of nitrogen monoxide (a) and nitrosyl chloride (b).

we refer to them as Type III metals in respect to the gas modulation effect.

Type II and III metals were tested for their activity against NO derivatives, namely, gaseous nitrosyl chloride (NOCl, Fig. 6(b)) formed *in-situ*. The action of NOCl on the Pt NPs and Au NPs was similar to that of NO on Type I metals, and a dispersion of the nanoparticles over the surface was observed. Therefore, we refer to Pt and Au as Type I* since their nanoparticles are still capable of rapid atomization. Silver also shows resistance to sputtering in this case, and the effect of nitrosyl chloride is not visually different from that of NO even when treated sequentially with NO, NOCl and NO (see the Supporting Information for more details). The metals Ir, Ru, and Rh are stable, and no motion, dispersion or growth of the nanoparticles is observed in the presence of nitrosyl chloride.

Analyzing the obtained results of the NO and CO effects in this work, as well as previous work on the effect of CO on Au nanoparticles [53,92], we suggest a key influence of the boundary orbitals on the RGM. Pd nanoparticles were dispersed by reaction with NO and were stabilized by interaction with CO, while gold particles were dispersed in CO atmosphere and agglomerate under NO atmosphere. From electronic reasons, NO is the electron density acceptor of the metal (Table S4), whereas CO is the donor, and the observed effects are opposite with respect to the mentioned metals. Thus, it can be assumed that in complex nanoparticle, support, and gas systems, electron density donation from CO occurs on an unoccupied orbital, which is bonding for Pd and anti-bonding for Au, resulting in enhanced bonding between metal atoms in the former case and weakening in the latter. When the electron density at NO is considered, the opposite pattern should be observed, namely, weakening of the bonding between palladium atoms and strengthening between gold atoms, which were in agreement with our work and previous works. The degree of atomization and aggregation should be determined by a complex factor depending on the ability of the ligand to donate or accept electron density, the initial bond strength between atoms in the metal, the distance between electron levels, which depends on the particle size, and the influence of the support, which is also capable of exerting electronic effects and both enhancing and counteracting the Reactive Gas Modulation.

4. Conclusions

This work demonstrates the concept of Reactive Gas Modulation (RGM) as a versatile tool for controlling the nucleation of metallic nanostructures on a support surface. It was found that exposure to NO, CO, H_2, O_2, NOCl and H_2O molecules leads to directional changes in the morphology and mobility of nanoparticles, including their disintegration to the atomic state or, on the contrary, aggregation.

Particular attention is paid to the action of NO, which effectively degrades palladium nanoparticles to the single-atom state, accompanied by a significant increase in catalytic activity in C–C bond cross-coupling reactions (*e.g.*, the Suzuki-Miyaura reaction), even under ultrasoft conditions and at low palladium content. It was concluded that NO-induced metal sputtering is facilitated by weakening of Pd–Pd interatomic bonds, reduction of migration barriers, and formation of active centers in the form of individual atoms.

The motion of a single atom on the graphite surface has an energy barrier of 41.1 meV, but NO coordination reduces this barrier to 9.6 meV, as well as for the considered Pd_{38} nanoparticle, where a decrease in the barrier from 0.39 to 0.23 eV was observed. In addition, the Crystal orbital Hamilton population analysis revealed the antibonding effect of NO at the Fermi level for palladium. COHP analysis revealed that in the case of a single Pd atom on graphene, the adsorption of NO promotes the appearance of antibonding states between Pd and graphene, leading to a decrease in the migration barrier. The adsorption of NO on the Pd surface leads to a weakening of the Pd–Pd bonds, as evidenced by the ICOHP analysis, resulting in a decrease in the bond strength and an increase in the Pd–Pd bond length. This may explain the effect of NO in breaking the strong Pd–Pd bond and, consequently, destroying the surface nanoparticles. The relatively high displacement barriers in the presence of H_2O and O_2 for the individual atom and the Pd_{38} particle, equal to 0.0705 and 0.0624 eV and 0.38 and 0.29 eV, respectively, are nevertheless overcome, which is reflected in the recorded motion of the nanoparticles. Oxide supports (SiO_2) stabilize palladium nanoparticles to a greater extent than graphite. The energy of barrier of a Pd_{38} nanoparticle motion in an NO atmosphere is 0.71 eV (versus 0.23 eV on graphite), leading to an increase in the number of undispersed nanoparticles vs graphite support. However, palladium atomization readily occurs in this case, as confirmed experimentally.

The combination of quantum-chemical calculations, electron microscopic analysis performed with nanometer precision before and after treatment with these gases, and machine learning allowed us to obtain a detailed view of the mechanisms of gas-induced rearrangement of metal particles. The obtained results are substantiated both at the level of individual atoms and at the scale of the nanoparticles, as well as confirmed in practical catalytic tests and were scalable to the gram level.

The concept of RGM introduced in this study opens a new path to the development of adaptive, highly efficient and stable catalysts based on single atoms. The simplicity of implementation, applicability to various metals (Pd, Ni, Fe, Co, Ag, Cu, Au, Pt, *etc.*), versatility of the support (both oxides and carbon materials) and the possibility of fine-tuning the catalyst activity make the proposed approach promising for both fundamental research and industrial applications in the field of sustainable organic synthesis and beyond.

Electronic supporting information

Supporting Information. Materials and methods, TEM images, SEM images, EDX-mapping, HAADF-STEM images, DFT calculation details, and optimization tables

Notes

The authors declare no competing financial interest.

Acknowledgments

The authors acknowledge Russian Science Foundation (23-13-00171).

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Graphical Abstract

Chin. J. Catal., 2026, 84: 324–336 doi: 10.1016/S1872-2067(26)65004-8

Reactive gas modulation alters metal nanostructures nuclearity and boosts catalytic activity

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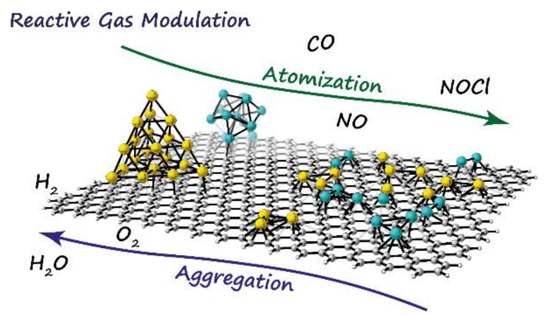

Treatment with a reactive gas allows control over the size and distribution of metal nanoparticles on the supports. NO fragments Pd nanoparticles into highly active individual atoms, significantly increasing catalytic activity in cross-coupling reactions.

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活性气体调控金属纳米结构核性并提升催化活性

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摘要：多相催化是现代工业的基石，涉及超过80%的工业催化过程，其优势在于催化剂的高耐久性、低产物污染以及易于分离．然而，与均相催化相比，多相催化剂的活性往往较低，尤其是在精细有机合成领域．在多相催化剂中，单原子催化剂(SACs)因其活性中心以原子级分散、催化行为接近均相体系而备受关注，在碳-碳交叉偶联等反应中展现出巨大潜力．然而，SACs的规模化应用仍面临合成方法复杂、难以实现纯单原子分散状态等挑战，实际反应过程中催化剂金属活性位点通常以单原子与纳米粒子共存的形式存在．

本文在温和条件(低压、低温)下，揭示了金属纳米粒子在活性气体(CO、NO、H_2、H_2O和O_2)调控下的表面动态行为．结合量子化学模拟、实验方法和机器学习方法，阐明了不同活性气体对金属纳米结构核性的调控机制：NO促进纳米粒子分裂成高活性的单原子物种；H_2、H_2O和O_2诱导纳米粒子生长；CO则起到稳定纳米颗粒的作用．基于活性气体调控效应，可灵活控制纳米颗粒的尺寸和分布，为金属纳米结构核性的调控提供了便捷路径．以NO气体处理的Pd/C催化剂为例，其在35 °C温和条件下显著促进Suzuki-Miyaura交叉偶联反应．此外，该方法对Ni、Fe、Co、Cu、Au、Pt、Ru、Ir、Rh等多种金属均表现出了适用性，展现出了广泛潜力．

综上，本文通过理论计算与实验相结合的多层次研究，开发了一种快速、节能、易于推广的活性气体调控方法，实现了克级规模SACs的可控制备，为精细有机合成中高效催化体系的设计提供了新策略，也为催化与材料科学在纳米尺度上的发展开辟了新路径．

关键词：单原子中心；金属纳米颗粒；动态表面现象；金属核性调控；催化剂活化；多相催化

收稿日期: 2025-09-09．接受日期: 2026-01-12．上网时间: 2026-05-05．
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基金来源：俄罗斯科学基金(23-13-00171).